

Strengthening GreenFuture BioChem's Sustainability: A Multifaceted Approach

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Project Objectives

This report will provide comprehensive recommendations for GreenFuture BioChem; a global manufacturer of bio-based specialty chemicals, to help them achieve their mission of replacing petrochemicals with sustainable alternatives across multiple industries with the end goal of finding the balance between sustainability and profitability. Our recommendations will be supported by our multifaceted approach which includes, our objectives informed by our assumptions, and our justifications of our recommendations through our integration and analysis.

Project Scope

GreenFuture BioChem is a global manufacturer of bio-based specialty chemicals, our goal is to help them achieve their mission to replace petrochemicals with sustainable alternatives across multiple industries. We were provided the company's portfolio which consists of 50 brand families (750 existing SKUs) serving five target markets: Automotive Solutions, Consumer & Home Care, Industrial Lubricants, Packaging Materials, and Specialty Polymers. We also considered how their products do include bio-based lubricants, biodegradable plastics, natural surfactants, and renewable packaging films. In terms of manufacturing operations, GreenFuture BioChem operates in Houston, USA; Frankfurt, Germany; Shanghai, China; São Paulo, Brazil; and Mumbai, India, sourcing its raw materials from suppliers across North America, Europe, Asia, and South America. GreenFuture BioChem's customers are from the aforementioned four regions, plus the Middle East and Africa. Being provided the following datasets: R&D Project Pipeline, Sales Pipeline, Manufacturing Production, Supply Chain Procurement Data, and Product and Brand Master Data Dictionary allowed us to organize, analyze and connect our assumptions to the company's current performance and mission.

To begin, we created an inventory from the provided datasets, to support our multifaceted approach, to ensure that multiple factors and variables are taken into consideration when finding ways to merge datasets and create assumptions to find the balance between sustainability and profitability.

Executive Summary

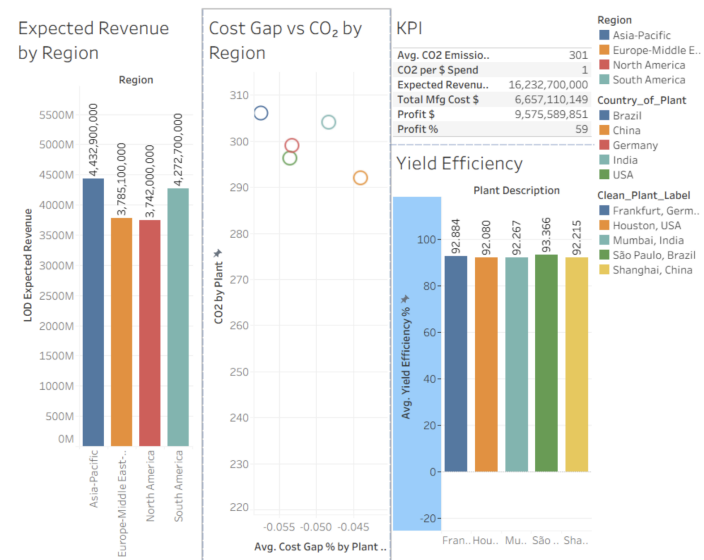
Considering the company's mission to replace petrochemicals with sustainable alternatives, operational data suggest that there will be implications for bottlenecks, risks and inefficiencies as sourcing bio-based chemicals are more expensive and their current manufacturing operations are limited to accessing these materials. Thus, to maintain sustainability and profitability the company must look into their current use of the cost of petrochemicals per region.

We can see a strong demonstration of innovation and efficiency with the following:

- 59% Profit Margin
- 93% Yield Efficiency
- 301kg/MT emission of CO₂

Key findings are:

- Great revenue being projected of \$16B dollars with the most coming from Asia Pacific Region
- Manufacturing efficiency, where US and China have higher Cost Gaps (difference between a product's or company's target cost and its estimated or actual cost)
- Project Conversion can improve, it is 86% (very good)
- For Sustainability, emissions are uneven across region



Data Methodology

The dashboards include information from all five provided datasets. These datasets included Manufacturing Production, Product Master, Product Metadata, RD Pipeline, Sales Pipeline and Supply Chain Procurement. We referenced our visual analytical platform: Tableau Prep which has a built in data cleaning resource. We merged all of the data sets in the following way. After inputting and merging together all of the datasets, they were cleaned and organized for us to begin our transformations, integration logic and assumptions. In order to expand the quality of data and gain a broader understanding of their key performance indicators we customly created the following; Yield efficiency percentage from manufacturing, expected revenue from sales, and CO₂ per dollars spent and total spend referencing the supply chain dataset. A limitation we came across was not being able to merge the supply chain to our master dashboard, thus having to navigate two different Tableau spreadsheets. Additionally, tableau does not offer the ability to share and co-work on dashboards together, thus we had to meet and show each other what progress had been made and make sure to stay on the same page, as well as time constraints to continue to elaborate our project and progress. Lastly, in terms of data quality issues, we did not come across any due to using Tableau Prep data optimizer.

Used Tableau to Clean and Prepare the Data through Data Interpreter. Added an additional Column next to Manufacturing to connect it to the Project.



Process of joining:

Manufacturing_Prudicts to SupplyChain_Procedure with the related fields of Country_of_Plant = County Origin

Product_Master to RD_Pipeline with the related fields of Category = Target Industry

RD_Pipleine to RD_Pipeline with the related fields of Project ID = ProjectID

Lastly, RD_Pipeline to Sales_Pipeline with the related fields of LinkedSalesOpp ID = SalesOpp ID

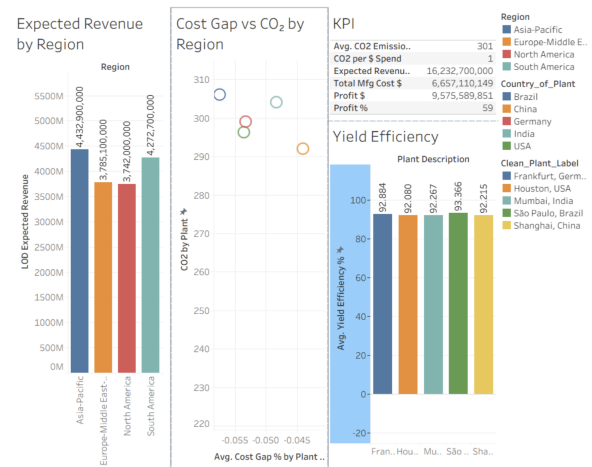
Assumptions

We approached our case study with one assumption: that all regions have their own natural unlimited resources and they do not rely on each other for the production of their units.

Analysis and Insights Supported by Dashboards

Considering the company's mission to replace petrochemicals with sustainable alternatives, operational data suggest that there will be implications for bottlenecks, risks and inefficiencies as sourcing bio-based chemicals are more expensive and their current manufacturing operations are limited to accessing these materials.

Dashboard 1-



Production is efficient & Consistent

Good Operational Profit

Nice Market Saturation

Asia Pacific seems to have High Growth Demand and Lower Raw Material Costs

Brazil is Cost Efficient

Regions with high CO₂ (example- Brazil) shows better Cost Control

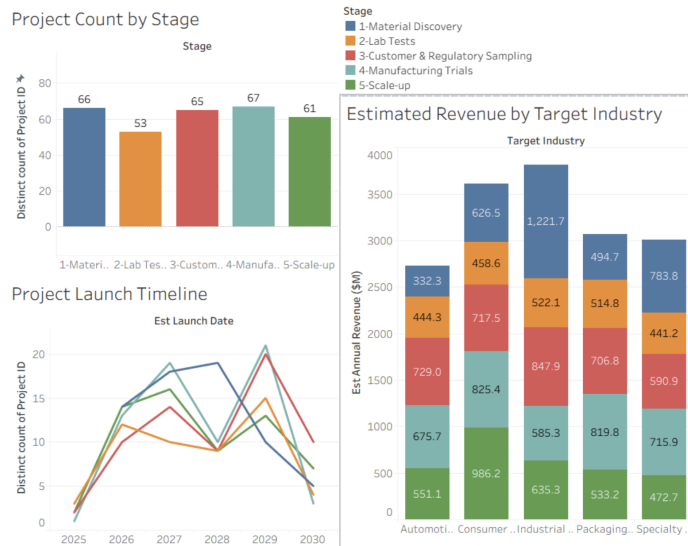
If its been made in Europe, it costs a little more but its

Europe achieves lower emissions at higher cost, reflecting a sustainability premium.

Strategic Implications-

- Shift cost-sensitive production from Europe to high-yield regions. This will increase the CO₂/unit
- Introduce a carbon-credit internal pricing model linking emissions; maybe set an internal carbon price (e.g., €40–€100 per ton CO₂) applied to every emitting activity.

Dashboard 2 -

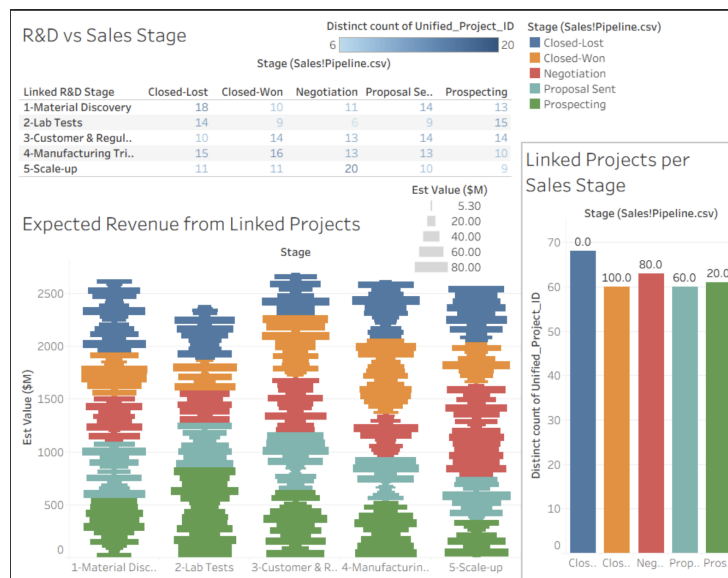


Two major R&D peaks in 2027 and 2029, each exceeding 18 projects.
 Noticeable drop after 2030, signaling a likely capacity or resourcing bottleneck.
 High volume of late-stage projects indicates strong portfolio discipline, but also heavier capital exposure.

Consumer & Packaging (\$3.87B) leads revenue potential.
 Automotive (\$2.5B) shows lower returns due to slower green material adoption.

Strategic Insights-

Prioritize investment in consumer and packaging R&D, where scalability and ESG demand are strongest.
 Shut down low-value or low-probability automotive R&D lines.



Strong Manufacturing Trials- Deals tend to close only after the product clears real manufacturing runs.

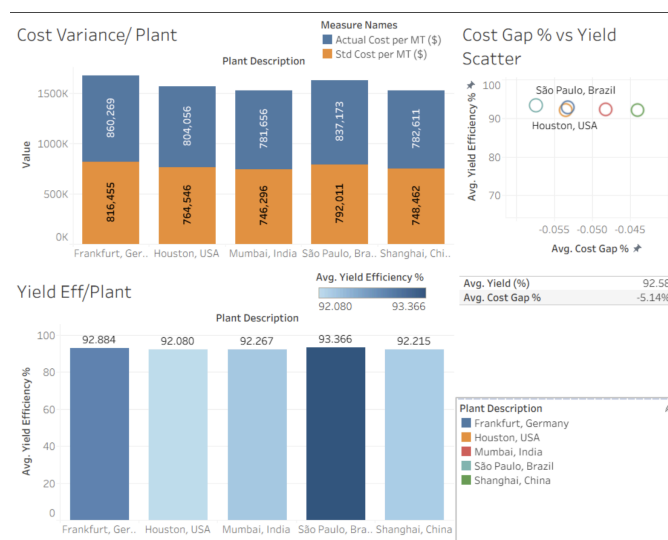
Weak Lab Trials- poor market fit signals during early development.

Commercial teams tend to join too late, leading to missed opportunities in early-stage projects.

Insights/Recommendations

- Track each projects maturity vs the likelihood of success. This may detect the weak stages while reporting as early as possible.

Dashboards 4



The *Cost Variance/Plant* chart shows cost overruns averaging 5 %, mostly driven by raw material inflation.

Brazil and Germany outperform in yield, while India offers the lowest cost per MT.

Actions we can take-

- Shift part of US & (Indonesia and Malaysia) output to Brazil & China respectively for better margins. These changes target systemic drivers (location, process, volatility) rather than surface-level cost cuts.

Dashboard 5

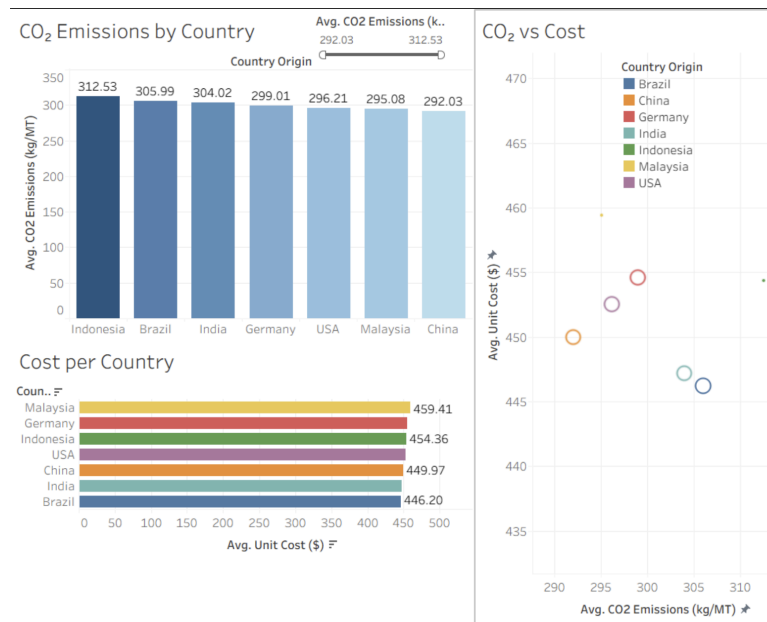


Figure 1.5

- Indonesia has the highest CO₂ emissions at 312.53 kg/mt
- Brazil is the second highest at 305.99 per kg/mt
- India is at an acceptable level
- Malasia has the best emission profile
- China is the most carbon efficient supplier

Co2 vs cost

- Malaysia have the highest bio premium product
- USA has a balanced profile
- China is the most efficient (lowest emission and most cost efficient)
- India has a moderate cost, it has a potential to supply more
- Brazil has low cost but has high emissions

From here we can say the lower cost suppliers often carry higher CO₂ emission/ exposure due to not having premium carbon efficient resources/clean energy

Malaysia and Germany pay sustainability premiums for using clean energy

Major Findings-

- We have a high Profit Margin (59%), this means we have a great yield stability and operational efficiency.
- Asia Pacific brings a lot of revenue but it is good to have mix of revenue coming from different regions (Market Diversity)
- CO2 Management remains globally uneven, this limits ESG credibility

Normal Findings from Dashboards-

- Cost Gap vs CO2: We can see that the cleaner regions face higher cost, this forces a trade off between sustainability & Profit
- The R&D launch waves show innovation cyclical, implying dependence on a few high-performing programs rather than continuous pipeline feeding.

ESG stands for Environmental, Social, and Governance, a set of criteria used by investors to evaluate a company's performance and sustainability. It assesses a company's impact on the environment, its relationships with people, and how it is managed. ESG is used in investing to guide socially conscious investment decisions and is also a framework for companies to operate more ethically and responsibly.

Sustainable and Actionable Recommendations and Summary

Recommendation 1: Consolidate Production Capacity Toward Brazil + Germany

Shift a portion of Houston (USA) and Mumbai (India) production volume to the São Paulo and Frankfurt plants, which have superior yield and stable cost structures.

- Brazil has 93.36% which is the highest globally according to the data. 1% improvement in yield will translate to millions in saving.
- Houston is more cost-volatile, not cost-efficient.

Impact

- Cuts average CO₂ by shifting away from USA & India
- If Yield stabilizes, Lower defects, and fewer reruns
- 2 to 3 annual reduction of CO₂

Risks

- Transition cost in shifting production workloads
- Brazil's energy mix is more carbon-intensive
- Time (1 to 2 years) for transition of operation

Recommendation 2: Rebalance/Shift the volume of Raw Material purchase away from Indonesia and Brazil toward China and USA suppliers to Reduce Cost and CO₂

- By shifting sourcing volume, the business simultaneously reduces cost per MT and CO₂ per MT
- China and USA are on the lower quadrant for Low Cost and Low Emissions
- Malaysia and Indonesia are on the upper quadrant for High Cost and High Emissions

Impact

- There will be a reduction in Raw Material Cost- non estimated 3%

- CO2 footprint reduction non estimated 3%
- Better ESG Rating (Environmental, Social, and Governance)

Risks

- Overdependence on Chinese Suppliers
- Limited Resources
- Extra Risks for geopolitical exposure
- Long term negotiations
- Time to negotiate with another supplier, that risks production, - limited raw materials- shortage in the market- not fulfilling demands and hence a risk of losing customers

Recommendation 3: R&D and Sales Development

Create cross functional teams for people working in Sales to early stages of Product Innovation instead of linking up at the end of the stage

- We have a lot of Closed won Deals
- A lot stuck in prospecting (Stage 1) which does not influence the Revenue
- Lack of commercial insight for Sales People

Impact

- Faster time to market which reduces time to market the product
- Lower failure rate in Scaling up a product
- High likelihood of hitting Projected Revenue

Risks

- Sales team may lack technical literacy initially
- More cross-functional meetings
- Time to integrate and come up with a plan (1 year)
- Training Cost, new better hiring

Recommendation 4: Maintain continuous discovery

- A lot of products launch until 2029 and drops in 2030.
- We should continue to increase early stage project funding and balance the R&D funnel
- This will sustain long term ecosystem and keep the market adaptable

Recommendation 5: Real time CO2 Dashboard

- Strengthens customer trust
- Connects Manufacturing emissions, Raw material emissions, Cost/ Region, Yield

Recommendation 6: Portfolio Rationalization Based on Target Industry Profitability

- Not all industries (Automotive, Consumer, Packaging, Specialty Chemicals) show equal performance. Recommending a shift in portfolio mix based on the stacked industry revenue plot could be a strategic move.

Declaration of AI:

We have used AI in summarizing some parts of its document and also doing level of detail for some of the calculations in Tableau.